Sentiment Analysis and Emotion Annotation of a Large-Scale Arabic YouTube Trauma Corpus

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***Abstract*—This study explores sentiments and emotions expressed in Arabic YouTube videos focusing on trauma narratives, offering insights into prevalent emotional responses and their implications for research, intervention, and support in the Arab world. Through meticulous data collection and analysis, sentiments and emotions were manually extracted from a sample of 25,025 comments within a dataset of 173,856 comments across 52 videos. Findings reveal diverse sentiments and emotions, with a notable prevalence of neutral sentiments and sadness. The study underscores the importance of culturally sensitive interventions and support mechanisms to address mental health challenges within digital discourse among Arabic-speaking YouTube users.**

Keywords—sentiment analysis, emotion annotation, YouTube videos, trauma corpus

I. Introduction

Psychological trauma occurs when an individual experiences an event or situation that they perceive as highly stressful. Any form of trauma can provoke stress reactions [1]. Typically, individuals express an initial sense of relief at surviving a traumatic event, but feelings such as stress, fear, and anger often follow this. Trauma may cause persistent thoughts about the event, and individuals may find themselves unable to stop thinking about it. In the age of digital media, online platforms have become a crucial space for individuals to share their stories and experiences, including those of trauma [2]. YouTube, one of the largest video-sharing platforms globally, hosts a wealth of content that reflects the multifaceted nature of human experiences, including traumatic events [3]. These videos, which are in Arabic and receive comments in Arabic, explore the emotional dimensions of trauma in the Arab world. Sentiment analysis and emotion annotation are powerful tools that allow researchers to analyze and understand the emotional content of large-scale datasets [4]. While these methodologies have been widely applied in English and other languages, their application to Arabic content, especially trauma narratives on YouTube, remains underexplored. Thus, a comprehensive study was conducted on sentiment analysis and emotion annotation of a large-scale Arabic YouTube trauma corpus. We seek to answer three key questions: 1. What common sentiments about trauma are expressed in Arabic YouTube videos? 2. What are the predominant emotions associated with trauma narratives in Arabic? 3. What are the implications of these findings for trauma research, intervention, and support in the Arab world?

By addressing these questions, our study will contribute to a deeper understanding of the emotional landscape of trauma in Arabic-speaking contexts through the sentiment analysis and emotion annotation of a large-scale Arabic YouTube trauma corpus. Moreover, it will inform the development of culturally sensitive tools and approaches for sentiment analysis and emotion annotation to enhance support and intervention for individuals affected by trauma in the Arab world.

II. Related Works

Sentiment Analysis is a computational approach that delves into people's opinions, attitudes, and emotions concerning a particular entity, whether individuals, events, or topics [5]. The sentiment is typically categorized as positive, negative, or neutral. In addition, researchers frequently conduct emotion annotation while annotating a dataset. Emotion annotation involves categorizing emotions into specific categories, such as the six emotions identified by Ekman: joy, fear, anger, disgust, sadness, and surprise [6]. These emotions serve as the foundational elements for understanding more complex emotional states. Analyzing sentiment across a large dataset can uncover trends and patterns in public opinion, which can be applied across various fields, including politics, marketing, and public health [7]. Sentiment analysis and emotion annotation have emerged as critical tools across a range of domains, particularly in the context of public health expression on social media [8,9,10,11].

These methodologies enable the computational analysis of individuals' opinions, attitudes, and emotional states, offering valuable insights into how emotions are expressed and perceived across diverse cultural contexts. This cross-cultural understanding is invaluable for research and developing more culturally sensitive and effective interventions and support systems. In the realm of mental health, sentiment analysis, and emotion annotation can help identify individuals at risk of mental health issues, monitor their emotional state over time, and provide early intervention or support [12], which is pertinent in the context of mental health, where cultural factors can significantly influence the expression and perception of emotions [13,14,15,16]. The ability to identify and monitor emotional states using sentiment analysis and emotion annotation can substantially enhance the quality of care and support offered to individuals with mental health concerns, contributing to improved mental health outcomes.

Sentiment analysis and emotion annotation have extended to studying the Arabic language. Applying these methodologies to Arabic content consists of both challenges and opportunities. There is a growing interest in analyzing social media output to understand public sentiment and trends in the context of Arabic social media. However, challenges such as the complex morphology of Arabic and the non-standard use of language in microblogs like Twitter, where tweets are often brief and informal, pose obstacles to practical opinion mining and sentiment analysis [17]. A separate study found that challenges in Arabic sentiment analysis include the complexity of the language, the need to translate figurative language accurately, a small community working on Arabic

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sentiment analysis, real-time analysis difficulties, the detection of figurative language for irony and sarcasm, identifying rhetorical questions, and dealing with synonymy and polysemy, all of which pose significant challenges for both human and machine analysis [18]. However, multiple studies in different fields have demonstrated the opportunity to apply sentiment analysis and emotion annotation in Arabic [19,20,21,22,23].

Recently, sentiment analysis and emotion annotation have been extended to mental illness, namely to the study of depression [24,25,26]. This expansion into mental health research underscores the potential for sentiment analysis to contribute to a deeper understanding of emotional states and their impact on well-being. As an outcome, the research has expanded to mental illnesses, such as post-traumatic stress disorder (PTSD) [27]. However, this differs from analyzing signals of PTSD [28] and the analysis of trauma [29]. The latter study explored Twitter's role in coronavirus disease-19 (COVID-19) support, analyzing support types, sentiments, and associations related to trauma/distress. Emotional support was sought most, with negative tweets prevailing, especially in early 2021, and it was determined that mask-wearing triggered trauma. Emotional support tweets were prominent, highlighting social support's crucial role in crises, which underscores insights for policymakers and health organizations to enhance pandemic response and aid individuals in need.

Trauma affects emotional expression and perception, which disrupts emotional regulation, causes hypervigilance and sensory overload, leads to emotional avoidance and difficulty trusting others, triggers re-experiencing symptoms, and fosters cognitive distortions. These challenges hinder individuals from effectively processing and communicating their feelings, necessitating trauma-informed interventions like therapy and support groups to rebuild a sense of safety, trust, and resilience [30,31,32,33]. While researchers have extensively studied PTSD and trauma , there are notable limitations, including a scarcity of sentiment analysis studies focusing on trauma and stressor-related disorders, and as of recently, there are no known studies on the annotation of trauma conducted in the Arabic language.

Sentiment analysis and emotion annotation represent promising avenues for comprehending and assisting individuals impacted by trauma across multiple dimensions. Through sentiment analysis, researchers and practitioners can discern recurrent emotional expressions among trauma survivors, facilitating the identification of triggers or stressors exacerbating trauma symptoms. By scrutinizing linguistic patterns and emotional cues, tools provide invaluable insights into the emotional journeys of individuals grappling with trauma. Emotion annotation enables the longitudinal tracking of emotional states, offering a mechanism to monitor fluctuations in mental health and well-being. By scrutinizing social media posts, text messages, and other communication channels, sentiment analysis can pinpoint shifts in mood or emotional distress, thereby enabling timely intervention and support for individuals susceptible to trauma-related manifestations.

Within the Arabic-speaking population, recent years have witnessed a surge in trauma attributed to the COVID-19 pandemic and other global crises. There has been a notable increase in social media expression. However, a paucity of research is dedicated to analyzing trauma in the Arabic language through social media, including sentiment and emotion annotations. Therefore, given the diverse cultural backgrounds of trauma survivors, it is essential to develop culturally sensitive tools and interventions.

III. Methodology

A. Data Collection

Psychological or emotional trauma refers to the lasting emotional and psychological effects resulting from distressing or overwhelming experiences involving a sense of threat to one's safety or well-being. These experiences can lead to a range of adverse reactions, including anxiety, depression, flashbacks, and difficulties in coping with daily life. We collected a wide range of terms related to trauma and stress from Arabic mental health sources to help us collect data on YouTube. Using YouTube for sentiment and emotion annotation is ideal due to its rich, diverse, and authentic user-generated content, multimodal data, large-scale availability, cultural and linguistic diversity, real-time and trending topics, engagement metrics, accessibility, and annotation tools. This allowed us to find, gather, organize, and filter comments systematically, resulting in a strong dataset that reflects experiences related to trauma and stress among Arabic-speaking YouTube users.

Step 1: Gathering Trauma and Stressor-Related Terms
To initiate the data collection process, we leveraged accessible Arabic mental health glossaries and online dictionaries [34,35,36,37]. This step aimed to compile a comprehensive list of terms associated with trauma and stressor-related disorders prevalent in the Arabic language.

Step 2: Identifying Relevant YouTube Videos
By utilizing the compiled list of trauma and stressor-related terms, we conducted searches on YouTube to identify the top videos addressing subjects pertinent to these terms. The objective was to pinpoint content potentially eliciting discussions, reactions, and sentiments related to trauma and stressor-related experiences.

Step 3: Selection of Target Videos
We focused on selecting YouTube videos with substantial engagement, specifically those with over 200 comments. This selection criterion yielded a total of 52 videos for further analysis.

Step 4: Comment Retrieval
To collect a sizable dataset for sentiment analysis and emotion annotation, we utilized the platform [communalytic.org] to facilitate the retrieval of comments associated with the chosen YouTube videos.

Step 5: Quantity of Retrieved Comments
The comment retrieval process resulted in the acquisition of a significant volume of data. We gathered 173,856 comments from the selected YouTube videos.

Step 6: Data Compilation and Preparation
We compiled the retrieved comments into an Excel spreadsheet. This compilation aimed to organize the data effectively, ensuring it was ready for annotation and analysis.

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Before manual annotation, we conducted a preliminary filtration process to remove any comments deemed incoherent or irrelevant to the study objectives. We sampled 25,025 comments from the dataset for detailed sentiment analysis and emotion annotation. This sampling strategy ensured a representative subset for comprehensive analysis while managing the annotation workload effectively.

B. Sentiment Analysis Approach

In our sentiment analysis of the Arabic YouTube comments about trauma, we employed a manual annotation approach by researchers skilled in both Arabic language and sentiment analysis techniques. Sentiment analysis involves identifying and categorizing the emotional tone expressed in text, including very positive, positive, very negative, negative, or neutral sentiments [38]. Additionally, the options of very positive or very negative sentiment demonstrate the maximum intensity of the sentiment. Each comment underwent careful examination to determine its predominant sentiment and was categorized accordingly. We relied on our expertise to identify emotional expressions within the Arabic text accurately, ensuring precise sentiment classification. Our understanding of trauma-related discussions in context allowed for detailed interpretation of sentiments, providing a comprehensive understanding of emotional responses among Arabic-speaking YouTube users toward trauma-related content. This meticulous manual annotation process facilitated thorough sentiment analysis and generated valuable insights into prevailing emotional attitudes and responses regarding traumatic experiences among Arabic-speaking YouTube users.

C. Emotion Annotation Approach

In our emotion annotation of the Arabic YouTube comments related to trauma, we annotated the comments using the Plutchik wheel of emotions, including joy, love, surprise, anger, sadness, fear, and disgust [39]. Each comment underwent an individual assessment to determine the predominant emotional state, categorized according to the emotions specified by the Plutchik wheel. This process involved careful examination of the text to accurately identify and classify the emotional content expressed within the Arabic language. By leveraging the Plutchik wheel of emotions, we aimed to capture a comprehensive range of emotional responses among Arabic-speaking YouTube users toward trauma-related content, providing valuable insights into the emotional landscape surrounding traumatic experiences in the Arabic-speaking community.

D. Inner Annotator Agreement (IAA)

We evaluated the inter-annotator agreement for sentiment analysis and emotion annotation in a sample of 750 comments from the YouTube Arabic trauma corpus, annotated by two psychology master's graduates, each with over 12 years of experience. Our results revealed that the sentiment analysis had a 75% agreement score, indicating a strong level of agreement between the annotators on the sentiments expressed in the comments. Likewise, the emotion annotation had an 88% agreement score, reflecting a high level of agreement regarding the emotional content of the comments.

We calculated Cohen's kappa coefficient to delve deeper into the agreement beyond simple percentages. For sentiment analysis, Cohen's kappa coefficient was approximately 0.689, indicating a substantial level of agreement between the annotators. For emotion annotation, Cohen's kappa coefficient was approximately 0.87, suggesting a high level of agreement. These results underscore the reliability of the annotations provided by the psychology master's graduates, validating the consistency and accuracy of their assessments.

IV. Results

A. Sentiment Analysis Results

The sentiment annotation of 25,025 YouTube comments provided insightful results, delineating the distribution of sentiments within the dataset. Negative sentiment comprised 23.6% of the comments, with 5913 instances, while neutral sentiment dominated, representing 47.04% of the dataset with 11,770 instances. Positive sentiment accounted for 22.46% of the comments (5624 instances), and very negative sentiment constituted 6.17% (1545 instances), with very positive sentiment being the smallest category at 0.69% (173 instances). These findings offer a comprehensive examination of sentiment distribution, highlighting the prevalence of neutral sentiments alongside notable instances of both positive and negative expressions.

Fig. 1 depicts the sentiment polarity visualization, showcasing the predominance of neutral sentiments among the manually annotated comments. The varied nature of neutral sentiments, encompassing discussions on mental health struggles, religious quotes, and responses devoid of discernible sentiment, underscores the complexity of online discourse.

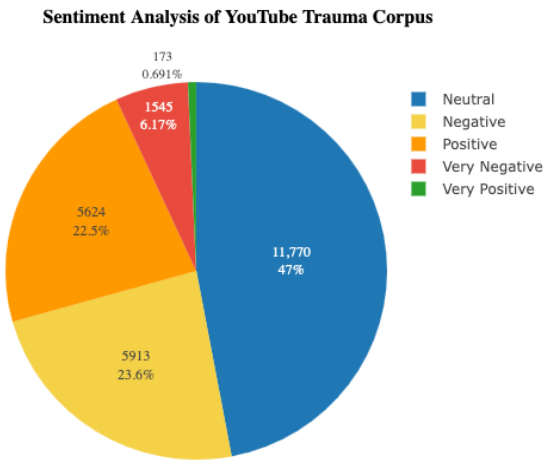

Fig. 1. Sentiment polarity visualization.

Table 1 illustrates the diverse expressions corresponding to each sentiment category, ranging from contemplation in neutral sentiments to expressions of resilience in positive sentiments and deep despair in negative sentiments.

Within the dataset, positive sentiment manifested in diverse expressions, from excitement to motivational messages, while negative sentiment encompassed feelings of hopelessness, sorrow, and worry. The example of positive sentiment demonstrates resilience despite adversity, while the

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negative sentiment example reflects a profound sense of exhaustion and self-despair. Based on the empirical evidence and sentiment analysis, this study elucidates the intricate spectrum prevalent in YouTube comments, underscoring the imperative of comprehending and mitigating diverse sentiments manifested within virtual social networks.

TABLE I. EXAMPLES OF NEUTRAL, POSITIVE, VERY POSITIVE, NEGATIVE, AND VERY NEGATIVE SENTIMENT ANALYSIS OF YOUTUBE COMMENTS.

Sentiment	Comment	Translation of Comment
Neutral	وهل يوجد سعادة فكيف تكون ؟؟	Was there happiness? How did it happen?
Positve	انا فخورة بقلبي تعرض لطعن والغدر والكسر والخذلان ومازال ينبض ...	I am proud of my heart that has been stabbed, betrayed, broken, and betrayed and is still beating...
Very Positive	الهي ربي يشفيج ويعافيج ويحقق حلمج ويطول عمرج ويبعد عنج كل مكروه وسوء يارب. 🤲💯	My God, may God heal you, grant you health, fulfill your dream, prolong your life, and keep away from you all harm and evil, O Lord. 🤲💯
Negative	تعبت من الايام ومن الدنيا ومن كل شئ حتي تعبت من نفسي..💔💔	I'm tired of the days, I'm tired of the world, I'm tired of everything, I'm tired of myself..💔💔
Very Negative	منهارة مكسورة تعبت نفسيا 💔💔😭▫️💯	Collapsed, broken, psychologically exhausted 💔💔😭▫️💯

B. Emotion Annotation Results

The emotion annotation of the trauma dataset provided valuable insights into the emotional content of the dataset. Most comments, comprising 47%, were classified as exhibiting no discernible emotion, totaling 11,770 instances. Sadness emerged as the most prevalent emotion, accounting for 26.2% of the comments, with 6566 instances. Love was the next most frequent emotion in 15.73% of the comments, with 3938 instances. Joy was identified in 7.3% of the comments, totaling 1830 instances. Fear, anger, surprise, and disgust were less frequently observed, with fear present in 0.92% (230 instances), anger in 1.77% (442 instances), surprise in 0.12% (30 instances), and disgust in 0.87% (219 instances) of the comments, respectively. These findings provide an understanding of the emotional spectrum present in the dataset, emphasizing the abundance of neutral content alongside notable instances of sadness and love.

Fig. 2 depicts the emotion annotation polarity visualization, showcasing the predominance of no discernible (none) emotions among the manually annotated comments. The varied nature of emotions with no discernible annotation reflects the examples stated in the sentiment analysis section. Examples in Table 2 illustrate the diverse expressions corresponding to each emotion category.

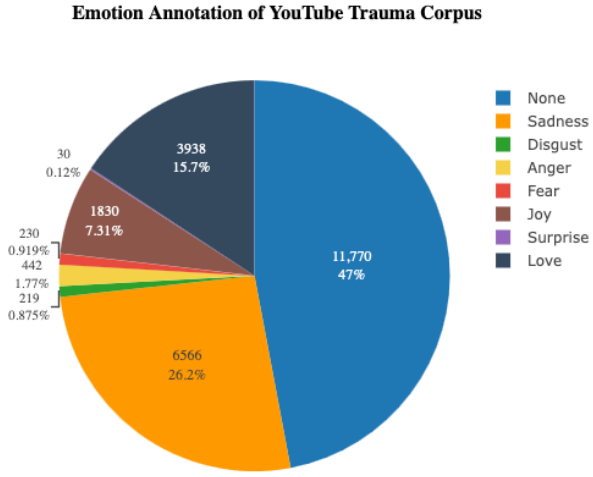

Fig. 2. Emotion annotation of YouTube trauma corpus.

In Table 2, we provide comments that offer a glimpse into a spectrum of emotions expressed within the dataset. One comment delves into the complexities of sadness, distinguishing between the weight of injustice and guilt. Our second example expresses love and goodwill towards an anonymous recipient, wishing them joy and encountering happy news. Joy is conveyed humorously in a message evoking laughter, while surprise emerges from a revelation challenging previous perceptions. Conversely, anger is palpable in a fervent invocation of divine punishment, while disgust prompts a plea for spiritual guidance. Fear is starkly illustrated in a distressing admission of imminent mortality, while profound sadness is articulated in a heartfelt prayer for solace and relief amidst brokenness. The comprehensive annotation of diverse emotional expressions in online interactions elucidates the intricate and multifaceted nature of human affective experiences, providing empirical insights into the complexity and depth of socio-emotional dynamics inherent in digital discourse.

TABLE II. DIVERSE EXPRESSIONS CORRESPOND TO EACH EMOTION CATEGORY.

Emotion	Comment	Translation of Comment
None	هناك نوعان من الحزن: يا نوع يصلط عليك الظلم من كل جانب وحزن يصلط عليك ذنب من كل جانب ان ظلمت	There are two types of sadness: one that afflicts you with injustice from every side and one that afflicts you with guilt from every side if you are wronged.

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Love	نحن لانعرف بعضنا ابداً ولكنني أحببت أن تتلقى رسالة من شخص عابر أتمنى لك يوما مليئ بالفرح والانشراح وأن يصادفك خبر سعيد يعيد لروحك الحياه قد يكون الله سخرني اليك حتى تبتسم أتمنى انا أكون نجحت في ذالك#يارب 🤍	We don't know each other at all, but I loved for you to receive a message from a passing person. I wish you a day full of joy and happiness, and that you encounter happy news that brings life back to your soul. God may have subjected me to you so that you would smile. I hope I succeeded in that. #Lord 🤍
Joy	هههههه يالله حسستني 😭😭 شريرة الله يوفقك يارب ويفتح □ دربك وينور مستقبلك	Hahaha, oh my God, you make me feel evil 😭😭 May God grant you success, open your path, and illuminate your future.
Surprise	بصراحه متلازمات القدس انت صدمتني فيها كان عندي افكار سلبيه بس مش سيئة ابدا بعد ما انت شرحتها	Honestly Jerusalem Syndrome, you shocked me I had negative thoughts But it's not bad at all after you explained it.
Anger	الله يجعلكم بنارر جهنم وبئس المصير اللهم آمين يارب العالمين اللهم أمين	May God put you in the fires of hell and an evil fate. Oh God, Amen, Lord of the worlds. Oh God, Amen
Disgust	اعود بالله كلامك بجيب مرظ نفسي استهدي بالله	I seek refuge in God. Your words cause mental illness; go seek guidance from God.
Fear	راح موت من الخوف والحزن وأنا فعمر 19 سنة	I am going to die of fear and sadness, and I am only 19 years old.
Sadness	انا جدا حزينة اللهم يارب جبرلي خاطري قد كسر قد ضاق صدري ياالله بالله	I am very sad. Oh, God, oh Lord, make me feel relief. My heart is broken. My chest is tight. Oh God, in God we trust.

C. IAA Results

The sentiment analysis of a sample of 750 comments revealed a 75% average agreement between annotators, highlighting a strong consensus on negative sentiments. The most significant agreement between the annotators was observed in identifying negative sentiment. However, discrepancies in agreement were noted, particularly regarding the intensity of negativity. Annotator one labeled comments as "very negative," whereas annotator two typically categorized them as "negative," with occasional discrepancies in the opposite direction. Similar discrepancies were observed in assessing the degree of positivity, where both annotators agreed on polarity but disagreed on its intensity.

The emotion annotation process yielded an impressive 88% agreement score. Despite categorizing sentiment before emotion annotation, discrepancies arose from annotators differing on the specific emotion expressed in the comments. Notably, we observed disagreements between annotators one and two regarding whether the emotion should be categorized as sadness or fear, anger or disgust, and love or joy. Our findings underscore the complexity of sentiment and emotion analysis, emphasizing the significance of meticulous interpretation in annotating emotional content.

Additionally, we computed Cohen's kappa coefficient to assess agreement beyond percentages. For sentiment analysis, it was approximately 0.689, indicating substantial agreement, while for emotion annotation, it was about 0.87, suggesting high consensus. Disagreements among annotators were thoroughly discussed and effectively reconciled. For sentiment analysis, the main discrepancy was not about the sentiment itself but its intensity. While the sentiment's polarity was consistent, the emotion annotation varied by perspective. To improve accuracy, annotators re-evaluated comments, leading to consensus on some labels. For example, anger was reclassified as disgust, and love was reclassified as joy.

In summary, while our analysis revealed a firm agreement on negative sentiments and impressive agreement in emotion annotation, discrepancies in intensity and specific emotion categorization underscore the complexity prevalent in sentiment and emotion analysis. Our findings highlight the importance of meticulous interpretation in annotating emotional content, as evidenced by the substantial agreement in sentiment analysis ($\kappa \approx 0.689$) and high agreement in emotion annotation ($\kappa \approx 0.87$) as assessed by Cohen's kappa coefficients.

V. Conclusions

Understanding the emotional responses within online discussions on trauma is crucial for developing effective interventions and support systems. Through sentiment analysis and emotion annotation, this study explored the common sentiments and predominant emotions expressed in Arabic YouTube videos related to trauma. The findings offer valuable insights into the emotional landscape of online communities in the Arab world and have significant implications for trauma research, intervention strategies, and support services.

Our research makes a significant contribution to the field by illuminating the intricate emotional dynamics within Arabic YouTube videos discussing trauma. We provide valuable insights into the prevalence of various sentiments and emotions, shedding light on the complexity of online discourse surrounding sensitive topics. Furthermore, our findings offer implications for trauma research, intervention strategies, and support services tailored to the Arab world's digital landscape. This contribution enhances our understanding of how individuals engage with and express emotions in online communities, particularly in contexts related to trauma, thereby advancing scholarship in digital communication and mental health.

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